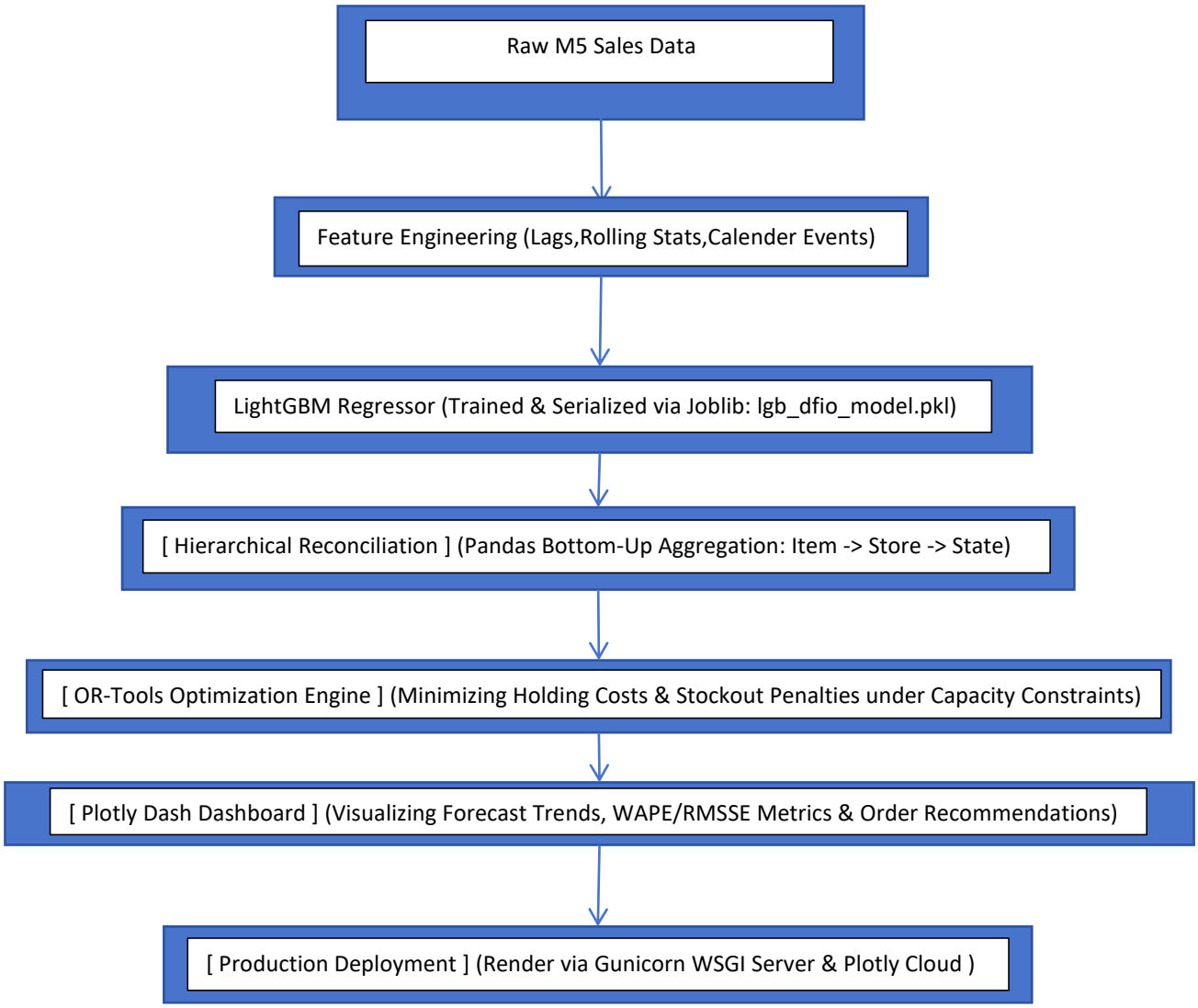
ARCHITECTURE DIAGRAM



API DOCUMENTATION

Component	Input Data Type	Output Data Type	Description
Model(lgb_dfio_model.pkl)	Pandas Dataframe(18 engineered features).	Numpy array(Point sales forecasts)	Generates item-store demand predictions.
Reconciliation Module	Granular item-store rows	Multi-level merged dataframe	Rolls up granular forecasts to store and state levels
OR-Tools Solver	Forecasted demand, inventory,cost weights	Optimal floats (Order quantity, Safety Stock)	Computes cost minimal replenishment quantities under constraints.
Dash UI	Serialized artifacts,static//dynamic tables.	HTML/JSON (Render&Plotly Cloud)	Renders the interactive visual client interface.